Ayush Patel

Materials Engineer

RESUME_1318

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Total Experience: 12 years

PROFESSIONAL SUMMARY

Materials Engineer with 12 years of experience in metallurgy, using Material Characterization, SEM, XRD, Metallography. Experienced in materials science, failure analysis and measurable delivery outcomes.

TECHNICAL SKILLS

Skills: DOE, Quality Control, XRD, Failure Analysis, Heat Treatment, ASTM Standards, Corrosion Testing, Material Characterization, Tensile Testing, SEM, Metallography

Methods: Materials science, Failure analysis, Quality assurance

PROFESSIONAL EXPERIENCE

Lead Materials Engineer | Ironleaf Engineering | Jan 2022 - Present

- Reduced material reject rate by 40% after heat-treatment window optimization
- Extended accelerated corrosion resistance by 37% in comparative studies
- Analyzed 194 metallurgical samples using documented acceptance criteria
- Coordinated engineering design reviews, acceptance criteria and test evidence with cross-functional stakeholders.

Materials Engineer | Evergrid Technologies | Jan 2018 - Dec 2021

- Analyzed 427 metallurgical samples using documented acceptance criteria
- Improved mechanical test repeatability by 36% through standard preparation

Associate Materials Engineer | VegaBridge Systems | Jul 2014 - Dec 2017

- Reduced material reject rate by 32% after heat-treatment window optimization
- Extended accelerated corrosion resistance by 40% in comparative studies

PROJECTS

Corrosion Resistance Trial | Corrosion Testing, ASTM Standards, XRD, SEM

- Tested coated coupons against accelerated exposure conditions.
- Documented acceptance criteria and evaluated the design using failure analysis and quality assurance.

Fracture Root Cause Review | Quality Control, XRD, DOE, Failure Analysis

- Prepared sample examination plans and failure mode documentation.
- Documented acceptance criteria and evaluated the design using failure analysis and quality assurance.

EDUCATION

B.Tech in Metallurgical and Materials Engineering, Horizon Institute of Engineering, 2014

Relevant Methods: Materials science, Failure analysis, Quality assurance

CERTIFICATIONS

ASTM Test Standards Workshop (2025)